

Why Shouldn't Sam Read? Toward a New Paradigm for Literacy and the Deaf

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Educators of the deaf and hard-of-hearing have long viewed this population as being Hearing people without the auditory sense and used instructional approaches, especially in reading, that conformed to or paralleled approaches taken with hearing children. In this article, it is argued that a paradigmatic shift must be undertaken that views deaf and hard-of-hearing individuals to be biologically, and therefore linguistically, different, yet uses visually based strategies for reading analogous or equivalent to those for hearing individuals. If these strategies are actively pursued and promoted by educators of the deaf and hard-of-hearing within the ASL/English bilingual/bicultural philosophy, it is suggested that the reconception of deaf and hard-of-hearing people as active, successful readers will be more easily attainable. These strategies, along with models of the reading process, are explored here to provide implications for instruction and areas of future research.

In his essay, "How Do You Dance Without Music?" (reprinted in Jacobs, 1989), the Deaf author Shanny Mow paints a compelling portrait of an "average" deaf person named Sam who relies on reading and writing for a significant portion of his daily routine (communicating with hearing co-workers, reading newspapers and books, etc.), yet encounters serious difficulties in maintaining access to the Hearing world through print, due to problems with vocabulary, idiomatic language, grammatical comprehension, and gaps in background knowledge. ("Hearing" is capitalized here to represent the community of persons with no significant hearing loss and their cultural/behavior norms and attitudes, to parallel the opposite usage of "Deaf." But "deaf"

and "hearing" are uncapitalized when relating to having or not having a hearing loss.) Mow's story, originally written in the 1970s, might seem quite dated and nonapplicable to a discussion of reading and the deaf or hard-of-hearing more than two decades later. Yet the sad truth remains that Sam's story could be equally true today as it was then: the average congenitally or postnatally deafened (before the age of two) child has long been recognized as achieving reading skills at or around the third to fourth grade (Allen, 1986; Quigley & Paul, 1989). Furthermore, this finding is not limited to the deaf population of the United States alone; studies of the results of the education of the deaf and hard-of-hearing in other countries such as Great Britain, Sweden, Denmark, and New Zealand have reached similar conclusions (Conrad, 1977, 1979; King & Quigley, 1985; Mahshie, 1995). Yet others, such as Lane and Baker (1974) and Moores (1990), have pointed out that there have always been cases of individual deaf and hard-of-hearing people who have surpassed these averages, reading at the twelfth or higher grade level. These findings indicate Sam's difficulties need not remain; therefore, how can we transform the majority of deaf people with reading difficulty into a minority? The first step lies in converting the mindset of low expectations into one of high expectations, which perceives deaf and hard-of-hearing people as potentially able, rather than incapable, readers.

Erting (1992) used Mow's story to ask: "Why can't Sam read?" This question, however well-intentioned, is the wrong one. The emphasis on the word "can't,"

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which connotes a physical or psychological inability, might suggest to the naive, or those already inclined to believe in a general inability of deaf persons to read, that the failure of deaf and hard-of-hearing persons to read with fluency is inevitable. Instead, researchers (and teachers) in the field of the education of the deaf should be asking, "Why *shouldn't* Sam be able to read?" By restating the basic question through the substitution of "can't" with "shouldn't," the emphasis is consequently placed not on an inability, but a potential (and even probable) ability. Such a shift in perspective can lead to a reconception of deaf and hard-of-hearing children and adults as individuals who are highly and actively engaged (or potentially so) in their constructions of literacy. Elsewhere, it has been suggested (Christensen, 1989; Israelite, Ewoldt, & Hoffmeister, 1992; Johnson, Liddell, & Erting, 1989) that most deaf and some hard-of-hearing people are best conceptualized as being linguistically different, not deficient, resulting from a need to learn primarily through a modality and language different from English, namely, American Sign Language (ASL). This argument will not be elaborated further here; however, I suggest that this linguistic and modality difference is also correlated with a visually based orientation to the world, which can include the use and appreciation of ASL, a visually based mode of communication. Although educators have traditionally viewed deaf and hard-of-hearing people as "Hearing" people with a "deficient" auditory sense (Branson & Miller, 1993; Lane, 1992), the thesis of this article will be that educators must instead identify and utilize visually based strategies for literacy development rather than those primarily rooted in sound, which is ultimately the weakest point of access to language and literacy for this population.

One might wonder why I am using the term "deaf and hard-of-hearing" in discussing the issue of reading in the population of those with a hearing loss, when the two are seemingly quite dissimilar populations. It is my contention that there are in actuality more similarities than differences between hard-of-hearing and deaf people in many areas of their lives, including literacy development. Although to be defined as hard-of-hearing varies widely according to the political, cultural, and audiological mores of the moment, it appears that most research has, taking a primarily audiological

orientation, defined hard-of-hearing as having hearing losses within or around the 25 to 79 dB range of loss (Grushkin, 1996). Hearing loss, even in the mild and moderate ranges, has a significant impact on speech production (syntax as well as articulation), as one might easily deduce, but more importantly, mild and moderate hearing losses (15–25 dB and higher) affect the academic performance (especially in the area of language skills) of hard-of-hearing children by as much as or more than a year, and these academic delays only increase with greater degrees of hearing loss (Quigley & Thomure, 1968; Ross, 1990; Ross, Brackett, & Maxon, 1982). Significantly, Ross (1990) suggested that hard-of-hearing children appear to be "bound by the literal meaning of words in much the same way as someone who is learning a second language" (p. 11). Ross's observation of hard-of-hearing children's linguistic performance in English is congruent with the arguments put forth by proponents of ASL/English bilingual education for deaf children, suggesting that hard-of-hearing children could benefit from ASL/English bilingual education, if one accepts the theoretical potential of this educational methodology for increasing proficiency in literacy for those with a hearing loss. There is some evidence that ASL/English bilingual education is a viable option for hard-of-hearing students: in a study of four hard-of-hearing children educated within an ASL/English bilingual/bicultural setting for deaf and hard-of-hearing children, Grushkin (1996) indicated that hard-of-hearing children do benefit from a visually accessible signing environment, while maintaining and improving their receptive and expressive English language patterns. Of course, this discussion is not suggesting that these areas of research apply to all hard-of-hearing (or deaf) individuals, but it is certainly true that there is a trend toward difficulty in areas of language and literacy for most hard-of-hearing (as well as deaf) children, especially with increasing degrees of hearing loss, and that both groups can derive benefits from coeducation within the same linguistic and academic setting, whether or not members of either group have learned ASL as a first or second language (Grushkin, 1996; Mahshie, 1995; Moschella, 1992).

As stated earlier, my central thesis is that research on reading in connection with deaf and hard-of-

hearing people must now turn on a paradigm of reading as a visually and cognitively rather than phonologically based process. Toward this purpose, the Goodman (1970, 1988) psycholinguistic model of reading will be endorsed, as this model deemphasizes phonological processes, concentrating on visual and cognitive processes instead. I present a brief overview of the critical importance of literacy for deaf and hard-of-hearing people and outline the several extant, competing models of reading. Following this, I discuss the reasons why a phonologically based approach to reading is not a sound one for deaf readers. Next, the evidence supporting a new, visual paradigm for deaf and hard-of-hearing readers will be submitted. Finally, I supply visually based instructional strategies derived from past and present research.

Importance of Literacy for the Deaf

For deaf children and adults, literacy is a vital means of accessing and learning about the larger hearing world around them. The ability to read and write is necessary in order to properly utilize newspapers, books, magazines, and other printed materials. In a survey of the reading habits of college-educated and noncollege-educated deaf adults (along with hearing college-educated students), Andrews (1978) found that the reading patterns between college-educated deaf and hearing persons were quite similar, as both groups read newspapers, magazines, and books for similar purposes and in similar amounts. Although the functions of reading for deaf and hard-of-hearing people are quite similar, it still remains that the necessity of literacy for maintaining contact with the world and others is more true for this population than it may be for hearing people. This is reflected in many of the technological innovations that provide printed representations of the spoken word in movies and television (closed captioning), telephone conversations (TTY/TDD), and the computer e-mail system and Internet, as well as the relatively widespread use of this technology by deaf and hard-of-hearing people. For successful acquisition and use of a language, it must have meaning and a purpose for the user; the realization that the failure to develop literacy skills will cut off one's enjoyment or use of these devices can be a powerfully motivating force in

promoting reading and writing skills in deaf and hard-of-hearing students.

Rottenberg and Searfoss (1992) asserted that for deaf children, literacy is used as a means of gaining entry into the Hearing world and learning about that world in three ways: as a primary form of communication, when they did not have the linguistic/lexical knowledge to express their interests; as a means of social interaction, through group writing, storytelling activities, and book sharing; and to make sense of the world, once they were aware that written language conveys messages. Conway (1985) was more specific about the uses of literacy for young deaf children. He identified both message-related and nonmessage-related purposes to writing. Message-related purposes included to preserve/recall experiences, convey personal information, organize general information, interact with a specific audience, and entertain oneself and others. While nonmessage-related purposes are not interactional, they are equally important for young children. Only two nonmessage-related functions were identified: to practice writing skills and to explore the mechanics of creating writing or written pieces. Thus, literacy does have many functions for deaf, as well as hearing children (Heath, 1983; Taylor & Dorsey-Gaines, 1988).

One important point to note is that for the deaf, literacy is almost entirely a visual process. While Hearing children may be exposed to literacy through visual means (reading and writing), they also have an auditory channel that can be utilized during storytelling periods. However, for most deaf individuals, even storytelling is conducted through the visual channel, through signs and/or lipreading. However, the main point remains in that literacy holds (or should hold) a more vital place in the life of the deaf and hard-of-hearing person than it does for the average Hearing person, due to the lack of (or diminished opportunity for) access to the multitude of alternative, auditory points of access to the world such as the radio, television, and telephone.

Toward a New Paradigm

Paradigms, or meta-theoretical models, are a set of beliefs about certain phenomena that serve to organize or drive people's perception or interpretation of these

phenomena. While paradigms in and of themselves may not necessarily be "wrong" (see Ritzer, 1992), it is certainly true that the application of a certain paradigmatic set can "blind" individuals to alternate explanations or perceptions of the same phenomena. Therefore, it can be said that paradigms *drive* researchers' interpretations of their work, but once a different paradigm or orientation to the same results is taken, the interpretations that were made may be seen in a much different light, by the same researchers or by others.

Operating at various times under paradigms of defect, deficit, disruption, and difference (Gormley & McGill-Franzen, 1978), research into reading for deaf and hard-of-hearing individuals has focused on physical problems ("defects") thought to be associated with deafness and reading, such as clumsiness (Beggs & Breslaw, 1982), or simply analyzed extant deficiencies in the written and spoken English comprehension and production of deaf and hard-of-hearing individuals (Clark, 1970; Ivimey, 1981; Klecan-Aker & Blondeau, 1990; Kluwin, 1979; Moores, 1970; Power & Quigley, 1973; Wilbur & Goodhart, 1985; Wilbur & Quigley, 1975). Alternative research has suggested educational failure has been at least partially a result of a changing population arising from social and technological shifts within society (disruption) such as a growing non-English-speaking population (Cohen, Fischgrund, & Redding, 1990; Cohen & Grace, 1990) and the increase of children with "multiple handicapping conditions" who might in earlier times not have survived to an educable age.

These foregoing paradigms have led to efforts to remove the "defect" or "deficit" through auditory amplification, intensive speech training, and drills in syntax and vocabulary. This approach, instituted in teacher training programs for teachers of the deaf, has emphasized oral speech and auditory training to the detriment of training in teaching skills, such as how to teach English or math (Coley & Bockmiller, 1980; Nover, 1993). Graduates of these teacher training programs have subsequently been poorly prepared to teach reading (Lane, Hoffmeister, & Bahan, 1996) and, given their predisposition to view the skills of deaf readers as deficient, to rely on poorly designed instructional materials such as basal readers and sentence-level syntactic instruction (Ewoldt, 1984; Israelite,

1988; Layton, Schmucker, & Holmes, 1979). Further, the reading lessons of deaf students have been characterized by frequent interruptions by the teacher to correct speech and reading errors, which in actuality provide the student with very little real reading time (Howarth, Wood, Griffiths, & Howarth, 1981; Limbrick, McNaughton, & Clay, 1992). Teachers are also led to view deaf students as unable to process complex materials, despite the students' own perceptions of enjoyment of more challenging materials (Ewoldt, Israelite, & Dodds, 1992).

More important, these foregoing orientations have been conducive to the creation of teaching methodology that viewed deaf and hard-of-hearing children as needing to *learn* (as opposed to *acquire*) "language" (which according to Nover, 1993, has traditionally been equated with English in both spoken and written forms), instead of viewing these children as being active learners with linguistic skills already in their possession. Toward this end, the difference model does not place the root of reading problems within the child himself or herself, favoring a view of the individual as possessing a language difference (as well as a language learning ability) that must be accommodated, if the child is to achieve reading proficiency. Hence, within the framework of this model, teachers are encouraged to view deaf children not as being "Hearing" people without the auditory sense, but as individuals with a language of their own that has a separate syntax, phonology, discourse structure, and even modality from that of English (even when they have not been exposed to this language from birth), which is, of course, ASL. Under the auspices of the difference model, in conjunction with the linguistic status of ASL, some educators of deaf and hard-of-hearing children have suggested that these children may actually learn English much in the manner of those learning a second language (Christensen, 1989; Israelite, Ewoldt, & Hoffmeister, 1992; Ross, 1990).

As a result (or precursor) of the difference model, much of the research conducted during the late 1970s through the 1980s appeared to have begun turning away from an English-based analysis of the written output of deaf children toward a recognition of the possible influence or transfer from ASL or other signed languages, drawing upon the research into second lan-

guage learning and acquisition of hearing people emerging at that time (Ivimey & Lachterman, 1980; Kluwin, 1979; Langston & Maxwell, 1988; Maxwell & Falick, 1992; Morariu & Bruning, 1984; Odom & Blanton, 1970). While this approach certainly bears further investigation, it is felt that a crucial element is missing in the investigation into the reading skills of deaf and hard-of-hearing readers. Some of the research cited above (Ivimey & Lachterman, 1980; Langston & Maxwell, 1988) has observed that deaf (and some hard-of-hearing) individuals appear to write in similar patterns, whether or not they had been exposed to signing, and, moreover, that they appear to process ASL in signed or written forms as equally appropriate forms of communication. If this population produces and accepts what has been termed "written ASL" regardless of whether they know ASL, there must be an alternative explanation for this phenomenon.

In an important study, Maxwell (1990) offered a possible partial explanation of the similarity of texts as written by signing and nonsigning deaf children, by looking to the basic biological needs of deaf children as a possible cause. Deaf (and some hard-of-hearing individuals), as a result of not being able to hear with any reliable acuity, must instead rely on their vision as a primary means of gaining information about the world around them. In examining individual children's signed narratives, she found them to possess many of the same elements of traditional signed narratives, and they were overwhelmingly centered on visual imagery. In one example, the children, retelling the plots of movies they had seen, produced images of what the movie scenes *looked like* through the use of storytelling, certain linguistic features of ASL, mime, and exaggeration. Maxwell pointed out that teachers, who are frequently unskilled in ASL, tended to attempt to reframe the child's discourse in English words, rather than as a set of visual images. This attempt at reframing causes many teachers to view the child's signing as "a bunch of structureless words" (Maxwell, 1990, p. 218).

Turning to the children's written narratives, Maxwell found that some children will write in a manner similar to their signed discourse; producing a visually centered narrative structure. In three texts—one with heavy ASL influence, one with little ASL influence (though still fairly ungrammatical in English), and one

with no ASL influence and a great deal of ungrammaticality in English—a common reliance on descriptions of people's actions (if not words), the setting, and visually accessible events (bombs, disappearances, etc.) was seen. Significantly, in the text with heavy ASL influence, the child described a dialogue with his or her mother, which was easily read by adult Deaf signers, who read it as a "partial transcription" of ASL, since the structure of the text appeared to consist of fairly alternating turns of dialogue, which can be read by competent signers through the use of "role shifting," a narrative discourse structure in ASL. The other texts, which described movies, did not illustrate internal motivations or dialogues, since without captioning (or the ability to read captions), these are inaccessible to deaf persons. Thus, as Maxwell stated, "Deaf Culture may have taken a *biological* propensity to relate to the world visually and developed it *rhetorically*" (1990, p. 227; italics in original). Although the question of whether deaf (and some hard-of-hearing) people are visually based, and whether this visual orientation translates into a need for a different model of reading and writing development based on visual processes, has been largely unexplored, it is certainly worth investigating.

Models of the Reading Process

As in almost every other field, research into how people read has spawned a number of competing theories that are hotly contested by their proponents. The two primary approaches are classified as being "bottom-up" and "top-down," referring to whether reading is seen as textually or cognitively driven. A third model, representing a compromise between the first two, perceives reading as an "Interactive" process. Like any other theoretical model, each of these approaches has strengths and drawbacks that explain or disconfirm aspects of the reading process in a wide variety of readers.

Bottom-up approaches constitute what people would recognize as the "traditional" model of reading, which emphasizes word recognition, either as whole word units ("sight vocabulary") or by analyzing grapheme-phoneme (letter-sound) relationships. The "phonics" approach, currently again in vogue, is one example of this traditional, bottom-up approach to reading in that it emphasizes reading as a process of

deciphering letters in written text into their component sounds. This is not to say that bottom-up theories do not encourage the use of other strategies; most of these bottom-up theories propose that a number of subskills also be taught to beginning readers, such as locating details, recognizing main ideas, pattern-matching, and so forth. Some bottom-up models of reading are more complex; for example, Gough (1972) posited a sequentially ordered set of mental transformations, as if the human reader was a computer with scanning abilities, that occurred in order for the reader to encode meaning from the text. Another model, developed by LaBerge and Samuels (1974), suggests that reading consists of three memory systems (visual, phonological, and semantic), which convert the input string into different formats leading to comprehension.

Bottom-up theories are so dubbed because they posit that reading begins at the page level (the most basic, "bottom" level), with cognitive processing resulting at the last step of the process. One criticism of the bottom-up models is that they are serial, one-way systems that do not adequately account for how some readers appear to bypass certain stages. More important, these models appear to treat reading primarily as a mechanistic, step-by-step process that does not allow readers to utilize their own knowledge and understanding of background information, contextual information in the text, semantic and syntactic knowledge, and so on. To illustrate, consider the first line of Lewis Carroll's "Jabberwocky": "Twas brillig, and the slithy toves did gyre and gimble in the wabe." Although one might "sound out" the words to come up with an appropriate pronunciation for these nonsensical words, bottom-up theories cannot adequately explain how we derive meaning from this text. Rumelhart (1994) recounts a number of studies that illustrate that processing at the semantic level can modify processing at the word level, lending further support to the criticisms of the bottom-up view of reading. One important criticism of phonologically based instruction offered by Goodman (1993, 1994) is that semantic relationships are maintained (in English and some other languages) at the cost of a sacrificing of phoneme-grapheme consistency. That is, dialectical and even phonemic differences within one speech event vary across regions and even within the individual speaker, none of which is commonly repre-

sented by standard written English. Further, it is well known that many spellings of English words bear little phonemic relationship to the actual pronunciation. In just one example, Goodman (1993) points out that the sound /n/ is variously represented by <kn> (knight), <gn> (sign), <pn> (pneumonia), yet even these are inconsistent (gnaw/ignite; sign/signal), rendering it difficult for young readers to dependably rely on letter-sound rules in reading.

In 1962, Kenneth Goodman observed that often children produced responses that did not conform to the text, yet these erroneous responses (or "miscues") were often not entirely inconsistent with the text, as nouns were replaced with other nouns, actions that were plausibly derived from previous actions were substituted for the action as written, and so on. These miscues led Goodman to consider readers as active, receptive users of language who engage in a "psycholinguistic guessing game" (Goodman, 1970), in which the reader relies on existing syntactic and semantic knowledge structures that minimize reliance on letter-sound relationships (which are inconsistently represented) within written text. While others use the term "decoding" to refer to the translation of letters and words into phonemic input ("sounding out"), Goodman calls this "recoding" due to the fact that "since print is a graphic code and speech is also a code, it is possible for readers to concentrate on matching print to sound with no meaning resulting. . . . [T]he readers go from code to code" (1988, p. 16). For this reason, Goodman reserves the term decoding to refer to the process of translating written text into a *meaning* code (Samuels & Kamil, 1988).

Since Goodman's research emphasizes the cognitive processes (the "top" level) that the reader brings to the reading task over the use of the written text as the driving force behind reading, his and the work of others in this vein have been labeled as top-down views of reading. Another, related top-down approach utilizes schema theory, which holds that the reading process is guided by a mapping of the textual input onto or against the existing schema (background knowledge or mental model) of the reader, and for the reading to be meaningful, all aspects of that schema must be compatible with the reader's input (Carrell & Eisterhold, 1988). Returning to the example of "Jabberwocky"

cited earlier, the psycholinguistic view would assert that although this line of text has only seven actual English words (out of a total of 13 words), readers are able to utilize their syntactical, contextual, and semantic knowledge in order to derive meaning from the text, which is the ultimate goal of reading. That is, readers can use the word "twas" to understand that "brillig" refers either to a time of day or a place, and that "sli-thy" is probably an adjective for "toves," which are some sort of animate creature, since they engage in the actions of gyring and gimbling in some sort of geographical feature termed "the wabe."

Like the bottom-up models, the top-down models have also been criticized on a number of fronts. They have also been accused of being too linear, emphasizing higher-level skills such as the prediction of meaning by using contextual clues or background knowledge at the expense of lower-level skills such as identifying lexical and grammatical forms (Eskey, 1988). In addition, it has been argued that these models may explain well the fluent, experienced reader, but do not explain the difficulties encountered by less proficient or developing readers. Eskey (1988) observes that some studies conflict with basic top-down tenets, such as that fluent readers are no more likely than poor ones to rely on orthographic or sentence context effects for the identification of words, and that poor readers are just as likely as good ones to rely on prior knowledge in deciphering texts, indicating that this strategy does not belong to good readers alone. Miscue analysis research itself, which provides much of the impetus for Goodman's theory, has come under fire for methodological and theoretical reasons (Leu, 1982; Wixson, 1979).

One important methodological criticism that Wixson (1979) raises is the question of varying linguistic competence among readers, which is also related to proficiency in reading. This issue is especially salient in the English as a Second Language (ESL) field. Devine (1988) observes that a number of studies has shown that linguistic competence is the best predictor of reading success in that second language, and that reading strategies, especially the ability to use various types of textual constraints, are related to the level of competence in that language. Grabe (1988) also points out that educators cannot assume that a large vocabulary or basic syntactic structures are already available for ESL

students, or that they possess the necessary relevant schemata for successful top-down processing of second-language texts. Although not directly related to the issue of linguistic competence (but pertinent to second-language readers), Grabe (1988) correctly identifies a number of important points: (1) second-language readers may not read in their first language; (2) even if second-language students possess literacy training, it is still not well-known how they approach reading (in their first or second languages) as a social phenomenon; (3) first-language reading abilities are often assumed to readily transfer to the second-language context, yet no strong evidence for this idea has been demonstrated; (4) second-language students coming from different orthographic traditions do appear to be affected by differing orthographic conventions, depending on their stage of reading skills acquisition; and (5) second-language readers do not begin reading English with the same English language knowledge possessed by English-speaking children. All of the points raised by ESL researchers, especially Grabe, have profound implications for deaf readers, and more so if one accepts the notion that deaf readers approach English in a similar manner to ESL learners.

Such criticisms of bottom-up and top-down models have led some to propose something of a compromise solution: an interactive model of reading, which recognizes both top-down and bottom-up processes, and suggests that reading is a process that consists of a continuous use or interchange of these two levels of processing. One notable example of the interactive theories is Rumelhart's (1994) model. Although this model initially appears linear, like the bottom-up models, Rumelhart makes an important contribution by showing how readers go back and forth between stages and levels of knowledge. For example, suppose a reader were to perceive a sentence as reading "The cat traveled at 60 miles per hour." Rumelhart suggests the reader might recognize that this interpretation is not consistent with the semantic and schematic knowledge of feline capabilities and reread the text to reidentify the "cat" as actually being a "car." Perhaps in response to previous criticisms of his research, Kenneth Goodman (1988) appears to have begun reformulating his conception of reading toward a more interactive stance, conceptualizing reading as a series of four "cycles" (op-

tical, perceptual, syntactic, and semantic) in which information is dealt with and meaning continually constructed. Goodman, like Rumelhart (1994), also emphasizes that readers may go back and forth between cyclical stages in order to reconfirm or correct constructed meanings that appear inconsistent with meanings or images presented earlier or later in the text.

It is important to consider how the approach to instruction of deaf and hard-of-hearing children generated by these theories of reading can be implemented, if at all, and the potential impact of a particular approach on the literacy development of these children. That is, bottom-up theories, with their emphasis on letter-sound correspondences, would seem fairly inappropriate for the majority of deaf and some hard-of-hearing children. Yet the psycholinguistic view presents problems as well, especially in the areas of concern expressed by second-language acquisition researchers, which are very similar to some of the concerns expressed by educators of the deaf. Interactive theories present a reasonable and promising balance between the two opposing viewpoints. Certainly, reading is a process in which the reader attempts to construct meaning from the text, but it is also just as true that the reader must recognize the physical symbols printed on the page, and somehow translate the aggregation of symbols into a form that constitutes meaning for the reader. Yet it seems to me that some interactive approaches also attend more heavily to the bottom-up aspects (such as decoding for letter-sound correspondences) than to the top-down aspects (predicting, utilizing background knowledge, etc.), which is not entirely applicable for those with significant degrees of hearing loss, for reasons to discussed in detail in the next section of this article. However, Goodman's modified (1988) theory, which views reading as an interaction between a set of cyclical phases (including physical perception) and emphasizes reading as a construction of meaning through visual, cognitive, and social processes rather than as a result (direct or not) of phonological processing, is congruent with my contentions that literacy for deaf (and some hard-of-hearing) individuals is accessed primarily through visual and cognitive processes. Further, it is suggested that this model

should be utilized in future research into reading development in deaf and hard-of-hearing populations.

Accessing Phonological Information Soundlessly?

Since at least the 1880s, with the rise of oralism after the Milan Conference of Educators of the Deaf (Lane, 1980), the role of audition and phonics instruction has been considered primary and even essential to the learning of English by deaf children. The quest for evidence of phonological recoding of written materials in deaf children has persisted, despite the apparent contradiction in terms of profoundly deaf individuals utilizing auditory-based approaches, and contradictory evidence for phonological strategy usage by deaf readers. I suggest that the search for a phonological basis in the reading skills of deaf individuals represents an ethnocentric perspective that is not entirely applicable or useful in the case of most deaf (and some hard-of-hearing) readers, due to the biological need of this population to relate to the world visually. Instead, I submit that "phonological recoding," if it exists within this population, should be viewed as an abstraction or recognition of visually based patterns of English acquired through lipreading and/or printed text. Yet phonological recoding should not be viewed as an absolute necessity for developing reading skills; deaf readers, like hearing readers, can and do use other strategies for word recognition that are not dependent on phoneme-grapheme relations.

For hearing children, some research points to a specific developmental hierarchy in acquiring literacy that favors a bottom-up view of phonological recoding as a requirement for developing higher-level or advanced reading skills. Mason (1980) identified three primary stages in reading: context dependent (reading only signs or labels within a familiar context, such as names of food brands, stores, or their own names); visual recognition (children begin learning to analyze words into letters and can read a few basic words); and letter-sound analysis (multisyllable words read independently, often utilizing sounding out processes). This basic developmental scheme has also been identified by others, with some variations. Ehri (1991)

adopted the terminology used by previous research to state that readers also go through three stages: the logographic, alphabetic, and orthographic stage. The first two stages are very similar to Mason's (1980) context-dependent and visual recognition stages, respectively. Yet, unlike Mason's letter-sound analysis phase, readers at the orthographic level of development learn to read similarly spelled sight words by storing alphabetic information about the words in memory, possibly through a hypothesized automation of the phonological recoding of letter sequences. Ehri felt this last stage to be the most advanced and advantageous for development of higher-level reading skills, since it minimizes the need for phonological recoding, which can create difficulties, especially with multisyllabic words (such as in "consignment"). Further, since words are read by sight rather than decoded letter by letter, and letters are identified much more quickly when they conform to familiar patterns, the reading process is speeded up since reading then becomes more a task of "automatic" word recognition rather than a decoding task. Reading at this level is highly consistent with Goodman's tenets, since he maintains that readers go directly from print to meaning. Further support for this viewpoint lies in a number of studies cited by Ehri and Wilce (1985), which indicate that even beginning readers at the first-grade level can process words visually without phonological mediation.

In addition, the case for the necessity of phonological recoding in hearing readers is not as solid as it may seem: Ehri (1991) notes that there are other ways to read words besides decoding and sight word reading. She states that new or relatively unfamiliar words may also be read by analogizing to known words as a whole (gave/have/brave, etc.) or as parts (investor = in/vest/or). Indeed, Ehri (1991) puts forth some evidence that reading by analogy is common among more advanced readers, but is a strategy that can be used by younger readers as well. Other strategies that readers can use are *analyzing and synthesizing orthographic patterns* (recognizing consistency in spelling patterns) and by processing contextual cues that enable readers to form expectations about words that lead to guesses as to what they are within the sentence or paragraph.

In the case of deaf readers, researchers such as

Conrad (1977) have suggested that the visual strategy is relatively inefficient in comparison to the phonological recoding strategy. Locke (1978), in a review of previous studies in this area, noted that these studies suggested that many deaf children were more closely oriented to graphemes as visual stimuli than to the phonemes they were intended to represent. In support of this idea, Locke noted that some of his subjects gave no evidence of inner speech, going from print to meaning through some nonspeech system, which was hypothesized as being either a shape code (as in a logographic strategy) or fingerspelling base. Although Locke (1978) suggested that children who exhibited "inner speech," silently pronouncing words as they read, were correlated with higher reading ability, he also suggested that models of reading, instead of emphasizing one strategy, should promote an interdependent framework in which syntactic, semantic, and phonological levels of processing are used. Locke's suggestion of an interdependent framework of syntactic, semantic, and phonological processing is quite reminiscent of the interactive models of reading, although his suggestion predates Rumelhart's (1994) framework by several years.

Kelly (1993) also found skilled, prelingually and profoundly deaf readers to possess a greater tendency to access phonological information when engaged in a speeded word reading task, yet he also discovered that recall/comprehension and processing time were not correlated highly with phonological facilitation, suggesting that this is not the only strategy that skilled deaf readers utilized. Kelly hypothesized that these skilled readers processed text in a manner which was more analytical than that of the average reader, in that they are more attentive to the conventions of English orthography, and that they maintain a more complete record of grammar during their reading, which is true of hearing readers as well. Thus, it would appear that the deaf readers in Kelly's study had progressed to the *orthographic stage of reading* described by Ehri (1991). It is important to note as well that the orthographic stage of reading and the behavior of the skilled readers described by Kelly are consistent with a visually based paradigm for reading in deaf and hard-of-hearing readers.

Quinn (1980) also found evidence for phonological encoding in her orally trained, severely to profoundly deaf and hearing subjects, yet additional findings indicated that the deaf might be reading less efficiently than the hearing groups, which might suggest that phonological encoding is a relatively inefficient strategy for this group of deaf readers. More important, Quinn appears to support Kelly's conjecture regarding the attention of skilled deaf readers to English orthography. Quinn discovered that deaf children, like the hearing adults in her study, appeared to be sensitive to contextual constraints in the texts, as well as processing the texts on a whole-word, rather than on a letter-by-letter basis, as was more characteristic of the hearing children in her study. A significant finding for Quinn was the fact that phonological recoding was not limited to the type of training the deaf children had received, as paradoxically, both orally trained and manually exposed children demonstrated evidence of this ability. In addition, Quinn observed the orally trained students in her study to be using reading strategies, such as chunking and other psycholinguistic processes, which have been found in hearing readers as well. The findings of deaf readers' use of context, whole-word reading, chunking, and other psycholinguistic strategies lends further support to Goodman's (1970) and Ewoldt's (1978) contention that the reading process is similar for both hearing and deaf readers, as well as to their support of reading as a holistic, meaning-based process.

However, it should be pointed out that although studies have frequently shown deaf people as a whole to display phonological recoding skills, as exemplified by Kelly (1993) and Quinn (1980), according to Gibbs (1989) there is no relationship between individual reading skill and reliance on phonological recoding in the 19 severely to profoundly deaf high school students (whose mean reading grade level was 6.7) in her study. Instead, the skilled readers were, as a group, found to be metacognitively sophisticated, demonstrating a high awareness of textual events and recognizing contextual errors and contradictions in comparison to the less-skilled readers. This metacognitive awareness is very similar to findings that reveal hearing readers also to be sensitive to context and cognitive foundations, as Goodman has maintained throughout his body of work.

Fischler (1984), while assessing recall for lists of letters that were similar phonologically, graphemically, or dactylally in profoundly deaf subjects reading at a sixth-grade level or better, who were asked to judge the meaningfulness of sentences (with or without a concurrent task intended to interfere with either articulatory or visual-spatial processing), found a longer decision latency rate for the sentence judgment task for deaf students in the visual interference condition, while latency for articulation and no task (control) conditions were virtually identical. Further, the visualization condition produced an interference effect to the same degree for the hearing subjects as it did for the deaf subjects. Fischler concluded from these results that hearing people use articulatory aspects in their processing, but the deaf do not. Instead, Fischler suggested that more skilled deaf readers, like hearing readers, are less reliant on surface representations of language than are less skilled readers, which is also consistent with the view of reading as a meaning-based task.

Treiman and Hirsh-Pasek (1983) also attempted to investigate whether second-generation, ASL-using Deaf readers use phonological recoding through a homophone rejection task. That is, given a sentence wherein a key word was homophonous with another, but semantically incorrect (e.g., "His favorite color is blew"), a longer decision time would indicate access of the phonological representation of the word. Consistent with the thesis I present, Treiman and Hirsh-Pasek's Deaf subjects did not take more time to reject the semantically incorrect homophones, indicating that they did not recode phonologically.

An interesting and significant aspect of Treiman and Hirsh-Pasek's study was their investigation of whether these Deaf subjects recoded into ASL (a visually based strategy) by showing them sentences in which several signs appeared similar on one or more parameters. As might be expected, the Deaf subjects made significantly more errors on sentences with similar signs, indicating that they did recode into ASL. Even so, Treiman and Hirsh-Pasek (1983) warned that sign recoding may be a less efficient strategy than phonological recoding, since sign recoding does not take advantage of the structure inherent in English syntax and orthography. Yet they gave no support for why this

might be so, despite this notion's running directly counter to the fact that deaf children of Deaf parents consistently demonstrate better reading skills than their peers educated through signed English and oral means (Corson, 1973; Meadow, 1968), despite their apparent use of sign-based recoding. Further research is needed to determine the role sign-based recoding, if it occurs, plays in the reading process for deaf and hard-of-hearing readers, and how the different syntactical and phonological/modality systems of the two languages are mediated within the cognitive schema of the individual reader.

There are a number of problems, primarily methodological and theoretical, with most of the research into phonological encoding for deaf or hearing readers. For example, short-term memory tasks (such as the lists of letters used by Fischler) are of limited use in understanding the reading process, since reading typically relies on long-term retention of and access to semantic and syntactic knowledge that is not tapped by decontextualized tasks involving relatively meaningless information (thereby disallowing the construction of meaning as would occur during reading in everyday life), rendering the results of these studies of little use in understanding the reading process as it actually occurs. This is not to deny that short-term memory does not play a role in reading; the reader must visually scan the printed text and hold it in mind prior to the mental process of translating the text into reading. However, it is only within the retrieval of long-term knowledge, concepts, and linguistic understanding that the real process of obtaining meaning takes place.

Leybaert (1993) observed, quite correctly, that Ehri's (1991) taxonomy (which maintains the necessity of going through the alphabetic stage, which relies on letter-sound correspondences) has implications for deaf readers, if this population cannot in fact truly access phonological information. In an extensive review of the literature on the use of phonological codes in reading by deaf individuals, Leybaert (1993) attempted to answer the question of whether the alphabetic stage of reading is possible, or even necessary for the transition to the orthographic stage in deaf readers. However, in examining some of these studies, it would seem that the results, interpreted as measuring phonological awareness, may actually associate with other, unrelated

processes. For example, some studies employ the Stroop task, in which words are presented in color, while the subjects are asked to name the color of the ink while ignoring the words (e.g., RED printed in blue ink). While deaf individuals were found to demonstrate an interference effect, the suggestion that this indicates an uncontrolled access to the phonological representation of the word is probably incorrect; instead, this task likely more accurately accesses conceptual (color and word) versus linguistic (word alone) conflicts rather than phonological processing *per se*.

Leybaert (1993) also covered several studies by Hanson, including one by Hanson and Fowler (1987), in which orthographically similar, rhyming words (bribe/tribe) were compared with nonrhyming pairs that were also orthographically similar (clown/flown). Although 37% of the deaf participants demonstrated an effect for rhyme, suggesting a phonological effect, 50% exhibited an effect for nonrhyming pairs as well, suggesting they relied on orthographic rather than phonological information in reading.

Another study by Hanson (1986) was discussed in Leybaert's review. In this study, the subject had to decide whether a target letter was present in a letter sequence that had recently been presented. The letter sequences were either orthographically regular (REMOND, SIFLET) or irregular (RDENMO, EFLSTI). Deaf participants with good speech demonstrated an advantage in regular over irregular strings, and it was suggested that they had developed a sensitivity to spelling-to-sound regularities, which were related to their oral production skills. However, this conclusion is not consistent with other studies, which suggest that most good readers possess a well-developed internal sense of permissible letter sequences within English orthography, facilitating their processing of these nonsense words (Samuels & Kamil, 1988). That is, although the orthographically regular nonsense words possessed no meaning, it is quite likely their very orthographic regularity enabled the subject to compare (analogize) the word (or its parts) to other similar words in their mental lexicon, facilitating the retention of the spelling of the word in short-term memory for the task of recalling the presence of certain letters to a greater degree than would be the case for a nonorthographically regular word.

Leybaert (1993) concluded from her review that deaf readers are "not condemned to be logographic readers; it is possible for them to exploit the alphabetic principle" (p. 302). However, Leybaert (possibly due to the narrow focus of her work) did not address the point that deaf readers can and do progress to the orthographic stage, which, according to Ehri (1991), is the most advanced stage of reading. In any case, how can these somewhat contradictory results on the use of phonological recoding by deaf and hard-of-hearing individuals be reconciled? Dodd (1980) and Hanson (1991) have suggested that profoundly, congenitally deaf individuals may receive their information about the phonology of English through lipreading and speech training. Similarly, Johnson (1994) and Leybaert (1993) have suggested that although deaf people may not have auditory access to the phonological system of English (or at the least, an incomplete or distorted version of English's phonological system), they may acquire this knowledge through the facial or mouth movements accompanying many ASL signs, whether English-like (Davis, 1990; Johnson, 1994), "reduced" versions of English words (Davis, 1990), or slightly "distorted" pronunciations (Johnson, 1994). These reduced or distorted English mouthings are differentiated from "ASL mouthing" (Davis, 1990), in which the accompanying mouth movements are not based on English words, but provide grammatical information not always contained in the sign, such as "cha" used while describing large objects. Johnson's (1994) notions are congruent with those of Hanson and Fowler (1987), who suggested that the phonological strategies observed in some deaf readers are due to the intensive speech and lipreading training that deaf children have traditionally received, regardless of the communication orientation (oral, Total Communication, etc.) of their respective schools. However, it is important to recognize, as Leybaert (1993) points out, that "the perceptual gaps related to lipreading provoke spelling deficiencies" (p. 294). In addition to the information that may be engendered by lipreading, Leybaert also suggests that fingerspelling may also provide a means of acquiring "an appreciation of the phonological contrasts of the language" (p. 275). However, it is significant to observe that all these "sources of phonological information" are entirely visually based, derived

from a process of recognizing and identifying visually based patterns in print and mouth movements or lipreading. However, Leybaert (1993) and Hanson (1991) caution that we must not expect the phonological information that deaf individuals hold to be the same as that of a hearing person; instead, it may very well be that due to the reliance on reading and writing (and mouthings), in addition to the relative inefficiency of their auditory mechanisms, the phonological information that deaf individuals hold may be highly non-standard.

Thus, it seems fairly clear that deaf readers may utilize phonological information about English, but this information is likely highly nonstandard and/or distorted to some degree. Further, this phonological information for the most part is likely not acquired auditorily, but rather as an abstraction of visually based recognition and identification of certain patterns of English acquired through print and lipreading. Yet this does not mean that educators should concentrate on developing awareness of phonological patterns in their deaf (and hard-of-hearing) students; such a strategy would be to continue to teach to the students' weakest points, as deficit paradigm-based education has done. Indeed, Leybaert (1993) cites a study by Hanson and colleagues from which she concludes that "it is possible to force individuals to rely on phonological information by making the orthographic information irrelevant" (p. 299), yet she goes on to observe that in comparison to hearing subjects in this study, the deaf students' performance was much poorer than that of their hearing counterparts, suggesting that an emphasis on phonological awareness is a poor strategy for instruction. Indeed, it would appear that orthographic awareness as a visual pattern seems to be more important, or even relevant, for deaf readers than the phonological strategy, despite assertions from those such as Conrad (1977), who have asserted that a visually based strategy for reading is relatively inefficient in comparison to a phonologically based strategy. From the evidence presented here, it seems fairly clear that it is not the visually based strategy that is inefficient, but that deaf readers are being taught or encouraged to read through auditory-based processes, as well as English-only (Nover, 1993) methodology that neglects the deaf and hard-of-hearing population's propensity toward

being visually based learners who may approach English more successfully as second-language learners. Since some deaf and hard-of-hearing readers do attain grade-equivalent and grade-appropriate reading levels through the use of visually based strategies, it can hardly be said that the visually based strategy is relatively inefficient in comparison to a phonologically based strategy. Moreover, it might even be concluded from these studies, as well as Goodman's (1993) points about phonologically based approaches to reading, that the phonologically based strategy is the more unsound strategy for deaf (and even hearing) readers. However, the evidence for or against phonological strategies for deaf readers is inconclusive; this issue should be re-investigated with a visually based alternative strategy held in mind.

Visually Based Processes of Reading

The evidence for the use of phonological strategies by some, if not most deaf and hard-of-hearing readers is inconclusive, yet evidence still remains that these readers are capable of processing written text in a manner similar to their hearing peers on a number of levels, including the visual and orthographic (even in those studies purporting to demonstrate phonological processing). It is apparent that educators must turn their attention to visually based and cognitive strategies for reading instruction, since these are two of the areas in which deaf and hard-of-hearing readers possess no disadvantage in comparison to their hearing peers. Since deaf readers have been demonstrated to use a visually based, orthographic strategy in their reading, and the orthographic strategy depends on, among other things, the "acquisition of context-sensitive rules, as well as the retention of orthographic specifics of the words" (Leybaert, 1993, p. 271), it is not a far stretch to realize that the Goodman (1988, 1994) model of reading should be adopted for deaf and hard-of-hearing students, for this model emphasizes, among other points, just these areas. Further support of this viewpoint will be presented in this section.

At the most basic level, it should not come as a surprise that reading competence for deaf readers is facilitated by one's breadth of vocabulary development, just as it is for hearing readers (Backman, Bruck, Hebert, &

Seidenberg, 1984; Kelly, 1996; LaSasso & Davey, 1987; Paul, 1996). Indeed, Gates and Chase (1976) observed deaf students to possess spelling skills that were much more advanced than those of their normally hearing peers. They concluded from their study that the skill of deaf readers in spelling is due to their utilization of a visual strategy in word perception. Gates and Chase appear to have been on the right track, since deaf readers apparently are well aware of the orthographic and morphological components of written English.

Friedman and Gillooly (1975) lent support to this idea in their study of orthographically allowable and disallowed nonsense words presented to deaf and hearing students in three elementary grade levels. They found that both deaf and hearing subjects were influenced by orthographic structure from the earliest grades on. However, by the fourth grade, the deaf students performed better than the hearing students on the disallowable items, suggesting that the deaf subjects had developed some undefined skill or strategy allowing the recall of the unstructured stimulus items. Both Quinn (1981) and Hanson (1985) have corroborated these findings, as they observed deaf individuals to be attempting to regularize irregular strings of fingerspelled letters into more regular forms, such as in the irregular string f-t-e-r-n-a-p-s to the more regular f-e-r-n-a-p-s, a-f-t-e-r-n-a-p-s, or f-e-r-n-t-a-p-s.

Perhaps the most convincing piece of evidence that deaf readers attend to English orthography, which also provides insight into how they learn to spell, comes from observers such as Schleper (1994) and Maxwell (1986). They, among others, have noted the frequent use of rebus-like strategies used to remember spellings, such as Schleper's (1994) narrative of a boy who remembers how to spell "reduce" by remembering RED ICE, with the I substituted by a U. These rebus-like systems are clearly not based upon sound, but visual patterns of words, based upon smaller, more familiar words within larger words, or as Ehri (1991) would have it, that deaf people are also using the analogy strategy for word identification (and recall).

As these rebus-like strategies indicate, deaf readers are also very attentive to the morphological regularities of English script. Hirsh-Pasek and Freyd (1983a, 1983b) conducted several studies in which words were presented that were related in both meaning and visual

appearance (SHIP/SHIPMENT) and those related in visual appearance only (PIG/PIGMENT) and asked subjects to divide derivationally complex words into their component parts (DIVISION to DI and VISION). A third experiment examined the subjects' abilities to employ their productive knowledge of English morphological structure through a cloze procedure with several nonsense words provided for the completion of the sentence. They found in the first study that the deaf readers were not confused by the visual similarities of the words presented to them and, further, were very similar to hearing individuals in their conservative pattern of errors, being reluctant to circle word pairs actually related in meaning. In addition, phonological information did not appear to distract the deaf readers from abstracting the morphological information; typically the converse is true for hearing readers. The second experiment found the deaf subjects to achieve an 80% level of correctness on these problems, with spelling changes in which the spelling of the word was changed during the derivational process (FORTUNE/UNFORTUNATELY) to be the primary source of error. This indicates that deaf readers are significantly more sensitive to these spelling changes than are hearing readers and again are not utilizing phonological information to the degree that hearing individuals will. The third task found deaf and hearing readers to make more errors on derivational endings than on inflectional endings (those which add /s/ for plural, or /ing/ for progressive), which again will transform the visual format of the word, creating some confusion for those inclined to rely on the visual rather than phonological "shape" of the word. The findings that deaf readers have skills perhaps superior to those of their hearing peers in visual sensitivity and analysis of the morphology and orthography of English were further corroborated by Hirsh-Pasek and Freyd (1984) and Hanson and Wilkenfeld (1985).

Deaf readers are also aware of the structure of English at the sentence level. Marshall (1970), utilizing cloze procedures, found predictions for function words (which composed 60% of the deleted items) to be more accurate for all levels of contextual constraint than for more concrete words composing the content class of items, especially with increasing degrees of structural cohesiveness. These early findings have been corroborated

by later work (Fischler, 1985; McKnight, 1989). Although utilization of context is important for hearing readers (Ehri, 1991; Goodman, 1970, 1994), Fischler suggests that deaf readers may rely on contextual clues to a greater degree than hearing readers do. Similarly, research on the ability to draw inferences from textual passages (Brown & Brewer, 1996) indicates that deaf and hearing readers alike are capable of making use of inferencing skills, regardless of their particular level of reading proficiency. The ability to utilize context, inferencing, background knowledge, and pictorial clues has led to improved comprehension of structures long thought to be difficult for deaf and some hard-of-hearing readers, such as the passive voice (McGill-Franzen & Gormley, 1980; Nolen & Wilbur, 1985) and idiomatic and metaphorical language (Conley, 1976; Iran-Nejad, Ortony, & Rittenhouse, 1980; Rittenhouse, Morreau, & Iran-Nejad, 1981).

The results of this research on utilizing contextual clues reinforces Bryans's (1979) and Gormley and Geoffrion's (1981) criticisms of practices in deaf education, which emphasized word- and sentence-level instruction in English, as these effectively prevented students from employing their semantic and experiential knowledge in the reading process, as well as from seeing text and discourse as a whole, continuous entity. These criticisms of word- and sentence-level instruction in English are also reinforced by a number of researchers (Ewoldt, 1984; Israelite, 1988; Layton, Schmucker, & Holmes, 1979; Yurkowski & Ewoldt, 1986), who have all presented criticisms of basal and adapted reading materials that, ironically, increased the difficulty of reading comprehension, even as they were intended to simplify reading for deaf children through the control of vocabulary, sentence length, and complexity of sentence form. Deaf individuals are also able to detect and explain logical inconsistencies within a given text to a degree comparable to that of their hearing peers, when prepared with sufficient background knowledge about the content area, according to Thornton, Harper, Choguill, and Kulhavy (1989). However, Yurkowski and Ewoldt (1986) warn that semantic processing also depends on clear, well-written text and adequate background knowledge, which can be provided through concrete experiences, as well as reading syntactically simpler materials that can later lead to suc-

cess with more syntactically complex materials. Indeed, Schirmer (1993) notes that deaf readers (like hearing readers) are *more likely* to engage in "deep-level, active cognitive processing" when they encounter textual materials that do not completely confirm their prior expectations or mental schema of story and/or content. These studies on context, metaphor, and logical thinking in relation to written text indicate that simplification of reading materials is not only problematic but unnecessary. When deaf or hearing readers at all levels are provided with text that includes natural, contextual information, they will employ a number of strategies, as well as their own experiential knowledge in order to successfully process this material.

Interestingly, Schirmer (1993) states that 89% of those subjects who engaged in making elaborated predictions during the reading task also developed mental imagery during reading. The exercise of mental imagery during reading is highly consistent with my contentions of reading as primarily a meaning-based task (for deaf and hearing readers) as well as a visually based process for deaf and hard-of-hearing readers. Although the use of mental imagery in deaf readers does not appear to have been well-explored, this is an important area for future investigation, within a paradigm of reading as a visually and cognitively based process.

This discussion has suggested that deaf readers can and do process English text through visually, linguistically, and cognitively based processes similar or analogous to those of hearing readers. Indeed, Goodman (1970) and Ewoldt (1978) have asserted throughout their work that there is only *one* reading process common to all readers, regardless of hearing or linguistic status and differences in language and modality.

Visual and Cognitive Instructional Strategies

Up to this point, I have suggested that deaf readers should be approached and instructed through visually rather than auditorily based strategies and that the whole language approach, which is derived from Goodman's (1994) view of literacy as a social and transactional process between the reader, the text, and their environment, be utilized. The whole language method approaches reading from a whole word and textual standpoint that emphasizes reading for meaning in-

stead of forming connections between speech and print. Goodman's (1988) model, more than the others discussed earlier, deemphasizes speech-based processes in favor of cognitive and linguistic processes, while recognizing the visual processes inherent in the reading task. Therefore, I suggest that this model be utilized for the instruction of deaf and hard-of-hearing children. Strategies conforming to the visual and cognitive strengths of the Goodman model, especially those dealing with attention to orthography and context, will be discussed in this section.

An important component of the whole language philosophy is the *exposure* of children to printed texts. One argument for the use of ASL in educating the deaf and hard-of-hearing is the well-known fact that deaf children of Deaf parentage tend to outperform their peers of Hearing parentage on tests of reading and writing skills. Some have attributed the superiority of their competence in English to the visual access to language, along with a corresponding exposure to and understanding of a wide range of personal and social experiences that ASL provides. While this is certainly true, the use of ASL alone does not explain how children from these families are being prepared to interact with the printed word. It is very likely that these Deaf parents, recognizing their own and their children's visual orientation, take pains to ensure that this visual orientation is maintained throughout every aspect of communication, from interpersonal interactions to reading. In a study of a hard-of-hearing mother (who uses ASL) reading to her child, Andrews and Taylor (1987) identified 14 strategies (categorizable into four main classifications) the mother used during book-sharing times, which reinforces the notion that Deaf parents utilize and maintain visually based strategies geared toward their child's eventual connection to the printed word. Among other strategies, the mother often signed on the child's body or the pictures, and eye contact was maintained for the control of interactions. She expended a great deal of effort to confirm the child's understanding through explanations, relating ideas in the text to the child's background experiences, confirming or commenting on her child's observations or the text, or requesting some action from the child. The mother also focused the child's attention on the book and its content through pointing, placing her finger on a spe-

cific word, or touching her child's body to illustrate a concept in the book. Explicit labeling of words or pictures was used, as was direct reading from the text (accompanied by pointing to the text with signs), and prompting of the child to read (or possibly more accurately, to predict) a specific word. Finally, the mother engaged in questioning designed to expand the child's understanding and thinking about what the animals or objects were doing, or what purpose they served. All of the mother's activities were geared towards language and reading goals and the connection of manual signs to the printed word. In other words, she employed visually based strategies while reading with her child.

The strategies employed at home can also be utilized, to some degree, within the educational setting. Mather (1989) discovered that a Deaf teacher (a native signer) transmitted information from the text in a manner that conformed to the grammatical rules of ASL. While reading to her students, she adapted her signs to the actions depicted in the pictures rather than to the accompanying text. Sound-based concepts were translated to visually based ones, such as meowing to crying. The teacher frequently changed roles with the characters in the text, which is a characteristic of ASL narrative discourse. Significantly, the teacher emphasized the connection between signs and the text by utilizing "miniature signs," in which she held the book facing the students and signed directly on the page or picture. Thus, this teacher demonstrated visually based strategies during reading time as well.

Andrews and colleagues (Andrews, 1988; Andrews & Mason, 1986) emphasize the necessity of aiding young deaf children in making the connection between manual signs and print. One technique for sign and print association was suggested by Robbins (1983) and Stofen-Fisher and Lee (1989), who found the pictorial representation of signs corresponding to written text to aid in the comprehension of text for deaf students. While this is not a method of reading instruction intended or recommended for use in the higher grades, it may be a feasible technique for aiding young children in making the connection between signs and the printed word.

Another means of connecting signs to print is through the ASL summary technique proposed by Andrews, Winograd, and DeVille (1994), in which stories

are first signed in ASL to the students, either "live" or on videotape. Following the signed presentation, the students are given the same text in printed form. The student may also be asked to retell the story in his or her own words, using his or her preferred communication modality and style. The signed presentation of the story appears to enhance students' comprehension of the written material, by providing them with some background knowledge, according to Andrews et al. (1994). Questions (in sign or print) about the story may also be asked of the students, either before or after the story has been presented, in order to influence their analysis and evaluation of the material (Dowaliby, 1992).

Some advocates for ASL/ESL bilingual/bicultural instruction of deaf children submit the hypothesis that reading in English for deaf students must consist of a mental process of translation between ASL and English. Akamatsu and Armour (1987) described a successful intervention study that promoted the development of these translation skills in deaf high school students through overt comparisons of how ideas were signed and how they were written. The study involved the use of transcriptions of signs through "glosses" and self-revised drafts of their written productions. This technique allowed the students to compare, with teacher help, the differences between signed and written structures and to develop internal translation and editing skills. Neuroth-Gimbrone and Logiodice (1992) also experienced success in fostering reading and writing skills with high school students through a process of promoting metalinguistic awareness of the differences between ASL and English, translating through glosses and written English, textual organization skills (word maps and grids), videotaped ASL narration, and signed ASL translation of written English texts. An important aspect of Neuroth-Gimbrone and Logiodice's work is that they actively sought to develop and increase their students' vocabularies, which is directly correlated with competence in reading (Backman, Bruck, Hebert, & Seidenberg, 1984; Kelly, 1996; Paul, 1996), through a visually based device (word grids) enabling their pupils to "see" the shades of meaning between semantically related words (scared, frightened, horrified, petrified).

Although much has been made of deaf readers' attention to orthography, context, and use of background

knowledge, it is not entirely true that these skills and strategies are natural and unlearned. They can be actively taught; Satchwell (1993) employed several techniques to develop skills in the areas of inferring, predicting, analyzing, attending, associating, synthesizing, and monitoring. Other skills such as chunking of text (Cooley, 1981; Ewoldt, 1978; Quinn, 1981); utilizing context and background information (Andrews & Mason, 1991); using one's own dialect and peripheral field information (Ewoldt, 1978); interpreting, questioning, paraphrasing, and word solving (Livingston, 1991) are also teachable. Similarly, writing skills such as revising; organizing information for structure, content, and cohesion; and editing for grammar (Gormley & Sarachan-Deily, 1982, 1987) are also important areas for instruction.

Similarly, recognition and comprehension of orthographic patterns inherent in English should also be brought to students' conscious attention; doing so can only facilitate their metacognitive awareness about language (Gibbs, 1989; Neuroth-Gimbrone & Logiodice, 1992). Ehri (1991) describes a series of lessons developed by Henry (1988) that concentrates on just this area, through the instruction (to upper-grade elementary students) about word origins (Anglo-Saxon, Romance, Greek) and the morpheme patterns associated with each origin type. The knowledge of root words and affixes by historical origin is extremely helpful for highly skilled readers; it only stands to reason that instruction in this area holds promise for developing readers in improving their lexical and semantic base. For younger children, this orthographic awareness can be promoted simply by teaching about such concepts as the consonant-vowel alternation common in English. Teachers can also aid in the development of orthographical awareness through informal means, such as by playfully signing the morphemes of an English word individually rather than as a whole sign in the rebus-like manner described by Maxwell (1986) and Schleper (1994) during relatively informal discourse, or when helping students learn to spell new or unfamiliar words.

Researchers in ESL have criticized the psycholinguistic viewpoint of reading on the grounds that ESL students do not possess the same levels of linguistic competence in English as native English-speaking chil-

dren; (Devine, 1988; Eskey, 1988; Grabe, 1988), nor do they possess the necessary relevant schemata for successful top-down processing of second-language texts (Grabe, 1988). In order to compensate for these areas of relative weakness, these researchers have provided support for an interactive view of reading, in which bottom-up and top-down processes are explicitly taught or encouraged. For example, Carrell (1988) and Eskey and Grabe (1988) recommend preteaching in vocabulary (multiple meanings and cross-cultural differences in meaning), including semantic mapping, which was done by Neuroth-Gimbrone and Logiodice (1992) for deaf readers. In addition to vocabulary, it is also important to teach about the various cohesive devices (substitution, ellipsis, conjunction, lexical cohesion) contained in English grammar, in order to help students utilize knowledge about these devices to aid in the reading process (Carrell, 1988; Eskey & Grabe, 1988). Eskey and Grabe (1988) also recommend activities targeted at the enhancement of automatic word identification skills (to quickly distinguish between orthographically similar words like sea, see, sew, saw) and reading rate building (by learning and recognizing phrases on an automatic level).

The bottom-up activities described above cannot act alone; all of the ESL researchers also point to the necessity of building background knowledge, or an alternative schema (Carrell, 1988; Carrell & Eisterhold, 1988; Eskey & Grabe, 1988). This can occur through organized methods as well as less-structured means. Both Carrell (1988) and Eskey and Grabe (1988) support prereading or text-previewing strategies such as the old SQ3R (Survey, Question, Read, Recite, Review) technique, or the Language Experience Approach (Carrell & Eisterhold, 1988; Stauffer, 1979). Text mapping (selecting key content from an expository passage and representing it in a visual display (diagrams, boxes, flowcharts, etc.) is one means of helping readers to make sense of text. In addition, predicting skills can be explicitly taught (Carrell, 1988).

A major problem for deaf and second-language readers is the fear of encountering unfamiliar words or phrases in the text that hinder comprehension of the text as a whole. Eskey and Grabe (1988) state that "the best 'strategy' for dealing with an unknown word may . . . be to keep reading until the meaning of that word

begins to make itself plain in relation to the larger context provided by the developing discourse as a whole" (p. 235). One means of helping students to understand that it is not productive to panic at the first unknown word or phrase is a technique cited by Carrell (1988) involving the use of texts on familiar topics in which anomalous words, phrases, and sentences are embedded. Students should be asked to stop reading when they encounter something that does not make sense, and a discussion of the anomalies and why they do not make sense helps to sensitize students to the importance of using background knowledge and comparing details of the text with their background knowledge. Texts with anomalous or nonsensical elements can be created by the teacher, or taken from other sources such as Lewis Carroll's "Jabberwocky" or Burgess's *A Clockwork Orange* (Carrell, 1988). However, it is also important to ensure that ESL (or deaf) students do not overly rely on top-down processing; for those who do, techniques are available to teach the importance of paying attention to textual details, such as explicit textual clues in one part of a text to distinguish appropriate from inappropriate possibilities in another area of the text (Carrell, 1988).

Related to the issue of anomalous or difficult language is the problem of utilizing and understanding figurative language, which is also true for ESL students and young children (Bisanz & Voss, 1981; Carrell, 1988; Carrell & Eisterhold, 1988). Although it has been demonstrated by Iran-Nejad, Ortony, and Rittenhouse (1980) and Rittenhouse, Morreau, and Iran-Nejad (1981) that provision of background knowledge and alternative schema is helpful for comprehension of metaphorical and figurative language, it must be noted here that ASL does possess metaphorical and figurative language (see Grushkin, in press; Klima & Bellugi, 1979). Comparisons of signed and printed uses of nonliteral language is a potentially useful, though largely untapped, means of bringing the students' understanding and knowledge of figurative language to the fore, and further research should be conducted in this area as well, especially in connection with the visual imagery that figurative language often brings to mind.

The Deaf community is of necessity a bilingual community (Grosjean, 1992). One aspect of this bilingualism, which is unique to the Deaf community, di-

rectly impinges upon literacy, and presents a potential advantage in developing reading skills for children of deaf and signing adults, is fingerspelling. Fingerspelling, which is a manual representation of English orthography, is an integral component of ASL typically used to convey proper nouns and English words and ideas or phrases that have no signed lexical equivalent in ASL, or simply to add emphasis to a statement (Padden, 1991; Smith, Lentz, & Mikos, 1988). Since fingerspelling, like the letters of the alphabet, is a means of encoding spoken (or written) English words, it has a great, if largely unrecognized or untapped, potential for promoting a direct connection between the printed word and the visual-spatial language of deaf and hard-of-hearing individuals. Indeed, deaf children have historically been found to possess greater skills in spelling English words than hearing children in comparison to their relative reading skills (Gates & Chase, 1976), and there appears to be a strong correlation between fingerspelling, vocabulary development, and general academic achievement (Quigley & Frisina, 1961).

Fingerspelling follows a fairly specific developmental pattern, according to Padden (1991), that appears to correlate with the progression of recognition of written language (Ehri, 1991; Mason, 1980) from holistic (logographic) to analytic (orthographic) processing. Padden (1990) says that fingerspelling is analogous to practicing written literacy skills for the deaf child; fingerspelling helps the child to form links between the language that he or she uses in everyday life and the characters that must be written on a piece of paper. However, the connection between fingerspelled handshapes and printed alphabetic letters does not seem to be made until around three years of age (Padden, 1991). Yet in several case studies, fingerspelling has been demonstrated to be a link between print and sign, enabling many deaf (and hearing children of deaf parentage) children to enter school with at least some basic reading skills (Soderbergh, 1975; Vernon & Coley, 1978). It is important to reiterate here, however, that fingerspelling does not provide a link to the phonology (sound system) of English, but rather to the orthographical system of print, as both are visually based systems providing a means of encoding spoken English concepts.

It should also be emphasized here that deaf chil-

dren do not automatically recognize the connection between fingerspelling and print; Hirsh-Pasek and Freyd (1983a), during their study of the utility of fingerspelling as a recoding mechanism, note that many of their subjects (6 to 11 years old) were "stunned" to discover a connection between something they had used all their lives (fingerspelling) and the printed word. However, it appears fairly clear that this connection should be emphasized fairly early. Hirsh-Pasek (1986, 1987) has found that deaf students tend to recognize a significant number of words in fingerspelling, but not as many are recognized in print. Yet, when these individuals are encouraged to decode the printed word into fingerspelling, their printed word recognition increases to the level of their fingerspelled vocabulary. Indeed, Hirsh-Pasek (1986) notes that the subjects in her study knew approximately three more words in fingerspelling than in print, yet never thought to use fingerspelling on the printed words until that strategy was suggested. Thus, it can be seen that fingerspelling can be used in a two-way process for literacy: fingerspelling to print and print to fingerspelling.

This discussion does not suggest that fingerspelling be utilized as the sole means of communication and instruction for deaf and hard-of-hearing children; this would be too unwieldy and inefficient, especially as a vehicle for the transmission of English skills, as Reich and Bick (1977) found. Instead, fingerspelling should be used in a naturalistic manner, according to the norms established by the Deaf community described by Padden (1991). That is, fingerspelling can be used for concepts or words that have no signed lexical equivalent, or when introducing a new English word (after giving the sign). It may even be possible to give a spelling test while using fingerspelled vocabulary. For example, it has long been recognized that some fingerspelled words undergo a lexicalization process (Battison, 1978) in which the fingerspelling becomes more sign-like and fewer elements of the actual fingerspelled words are used. As an illustration, suppose a teacher included the word "aquarium" as a vocabulary item. Initially introducing the word through fingerspelling (along with its printed representation), the teacher might initially and several times thereafter fingerspell the word quite slowly, allowing the students to visually retain all the elements of the fingerspelled

word. After several exposures, the teacher might gradually increase the rate of his or her fingerspelling to the normal speed that Deaf adults use. By this time, however, the students should have had sufficient exposures to the lexical item in fingerspelling and print that they would recognize the lexicalized version as the same word that they had been learning, which could then be used during a spelling test. Since, with lexicalized fingerspelling, individuals do not see all the elements of the actual word, the few elements that they do perceive would allow them (in conjunction with their memory of the actual spelling) to engage in a process of sounding out the word analogous to hearing students' retention of perceived phonemes in a spoken word (in conjunction with their memory of the word's spelling) to aid in their derivation of the correct spelling. It is not known if such a technique has actually been put into practice; however, this technique certainly bears investigation.

ASL and fingerspelling have been seen to play a role in children's writing development in observations of invented spellings by Schleper (1994) and Fok, van Hoek, Klima, and Bellugi (1991). Just as hearing children rely on phonology, so too do some deaf children utilize the signed phonological system. That is, these children will base their invented spellings or writings based on the handshape (or even the movement) used in the sign. Thus, "cat," which is signed with an "F" handshape, might be spelled C-A-F, but not K-A-T (Schleper, 1994), while "pie" might be "written" in a manner resembling the sign, which is presumed to be iconically representing the action of slicing a pie (Fok et al., 1991). Other students have drawn signs on paper along with writing when they did not know the spelling of the word or concept they wished to represent (Schleper, 1994). Even older students have utilized this strategy, according to Schleper, who told of a student writing "RRRR" to mean "keep your fingers crossed" (crossed index and middle fingers form the fingerspelled letter R in ASL). Further research must be done on how to incorporate ASL and fingerspelling within a whole language context.

However, none of the foregoing areas of instruction can be effective if deaf (and hearing) readers are not provided with actual time to *practice* their reading skills. This practice can take the form of guided, structured

exercises such as learning and reading new, similarly spelled words (pots, post, spot, stop), which help beginning readers to understand that letter arrangement (orthography) is important (Ehri, 1991). In addition, simply providing students with structured or unstructured reading time in order to explore different reading materials and practice on one's own skills learned during class (Howarth, Wood, Griffiths, & Howarth, 1981; Limbrick, McNaughton, & Clay, 1992) can be invaluable in promoting literacy development.

Another interactively and immediately relevant tool to employ in fostering the written literacy of deaf individuals is the TTY, which can be introduced at early ages (Neeley, 1994) as well as at older ages (Lieberth, 1988). A modern alternative to the TTY for fostering literacy through interactive text is the computer, which can be linked to other terminals through which students can converse with each other via their own keyboards and screen, such as the ENFI-style projects described by Marlatt (1996) and Peyton (1991), or through regular use of e-mail and chat rooms (which allow for real-time, alternating conversation) found on some on-line services. Although the use of these on-line services as a means of facilitating literacy has yet to be examined, it stands to reason that students (regardless of hearing status) would find it highly motivating to engage in activities that have a meaningful, interpersonal context through the computer, which is gaining an ever stronger foothold in modern society. However, it should be kept in mind that the conventions for writing (spelling, grammar, discourse structure, etc.) on the TTY (and possibly while writing informal missives through e-mail and while using chat rooms) are much different from those for other textual forms (essays, letters, etc.) in order to save time and money. Although literacy development through interactive media such as the TTY is important, it is also important to emphasize and distinguish to students the difference between writing styles and conventions for these media, and to know how and when to appropriately engage in one style over another in using these disparate media.

If teachers of the deaf employ (within a bilingual education context), or teach their charges to employ, the strategies outlined in this section that conform to

the visual-spatial needs of the deaf and hard-of-hearing, there is no reason why the majority of these students should not attain grade-level literacy skills. However, the strategies outlined here are relatively few; it is important for practitioners to develop (or disseminate) other visually based strategies that have been demonstrated to work for their classes.

Conclusions

It has been said that the pessimist sees a glass as half empty while the optimist perceives the same glass as half full. Here I have discussed how educators of the deaf and hard-of-hearing have traditionally turned their attention to deficiencies in the reading of deaf and hard-of-hearing individuals, rather than to their abilities, fostering a tradition of remediation rather than proactive instruction. That is, the focus should not be on what deaf and hard-of-hearing readers *cannot* do, but on what they *can* do. There is fairly ample evidence for a conception of deaf and hard-of-hearing people as being inherently biologically different, orienting themselves to the world through primarily visual rather than auditory means. Yet, despite this biological and even linguistic difference, deaf and hard-of-hearing readers employ strategies for obtaining meaning from written text that are analogous or equivalent to those of their hearing peers. Since it has been demonstrated that deaf and hard-of-hearing readers do employ strategies for reading that are similar to those of hearing readers, the Goodman psycholinguistic model of reading, which emphasizes that readers bring a wealth of semantic and syntactic knowledge to the reading task that enables them to glean meaning from text, is endorsed in this article. However, because deaf and hard-of-hearing readers evince many similarities to ESL students, concerns about the Goodman model for the ESL population apply equally well, for the most part, to deaf and hard-of-hearing readers. For this reason, some text-bound processing strategies should be taught explicitly in the classroom, and any necessary background knowledge or schemata that the deaf or hard-of-hearing reader may need in order to process the written text should be provided. In relation to this idea, instructional strategies aimed at raising the level of meta-

cognitive and metalinguistic awareness in students should be implemented, in order to further aid the students in bringing this knowledge to the reading (as well as the general academic) task. In addition, it is important that English grammar be explicitly taught, yet it is just as important that English grammar be taught within the context of ASL/English bilingual/bicultural education (wherein the grammar of ASL is also taught), since it is fairly clear that most deaf and some hard-of-hearing people, for the most part, do approach English in the manner of second-language learners. Instruction designed to impart knowledge about the phonology of English is not worthwhile for most deaf and hard-of-hearing readers (just as it is not for hearing readers), unless this instruction takes the form of encouraging these students to attend to this phonology as visually based patterns inherent within the English language, especially in the areas of spelling and morphology within common "sight" words.

Much of the research on reading and literacy has concentrated on what skills the deaf and hard-of-hearing are "lacking"; yet it has been found that when the research goes beyond a decontextualized, relatively meaningless level and provides the reader with a task in which syntactic and/or semantic knowledge can be drawn upon, deaf and hard-of-hearing readers have been found to demonstrate more reading skills than had been found previously. For this reason, it is important in future research to develop tasks that more closely resemble a "real" reading task, providing a meaningful, elaborated context in order to allow the reader to use the syntactic and semantic knowledge that he or she possesses.

There has been a great deal of interest in and research on the issue of contact between English and ASL in signing, but not as much exploring whether this contact arises in the written language of deaf and hard-of-hearing individuals. Future research should be directed toward the exploration of how the different modalities, phonologies, and representational systems of ASL and English come into contact, if they do, and how they are mediated in signs and writing. In connection with this issue, researchers should also investigate whether some deaf individuals utilize a sign-based recoding strategy for reading (instead of an auditorily

based strategy) and how the different phonologies and syntaxes of the two languages are accommodated (or possibly set aside) in order to write an English utterance.

It has long been known that English possesses a wide range of metaphoric and idiomatic language, which has presented some difficulties for deaf and hard-of-hearing readers. Yet there is very little understanding or knowledge of metaphorical and idiomatic language in ASL. Since one tenet of the bilingual/bicultural philosophy is that students must be made aware of similarities and differences between their two languages, future research should identify and compare figurative language in ASL in order to provide another area of linguistic comparison in English and language arts classrooms.

Earlier in this article, several points were made about second-language reading, such as the fact that second-language readers may not read in their first language, and how they approach reading as a social phenomenon is not well understood; in addition, there is some evidence that readers from different orthographic traditions are affected by the differing orthographic conventions (Grabe, 1988). In keeping with Grabe's points about biliteracy, investigations should be made about whether it is feasible or necessary to introduce or develop a writing system for ASL, which currently has none, with the possible exception of glosses, which are problematic for a number of reasons. Several systems for transcribing signs have been developed; one of these, SignWriting, has been implemented in several different countries, including the emerging Deaf community in Nicaragua, and appears to hold a great deal of utility and promise for these groups (Valerie Sutton, director of the Deaf Action Committee for SignWriting, personal communication, August 1996). The use of such a written transcription system for ASL remains largely unexplored for a number of reasons, but this area should be investigated further in the interests of developing a tradition of biliteracy within the American Deaf community. The use of fingerspelling, by educators and Deaf individuals, should also be further explored, since fingerspelling provides a potentially strong bridge to literacy for the Deaf community.

If the paradigm of linguistic and cultural difference

is adopted, which views the deaf individual as a person who must utilize a language with a different syntactic structure and even modality from that of English, every preestablished notion about the deaf is invalidated. Deaf people instead become second-language learners, with a culturally and biologically determined visually based way of perceiving the world, which may be reflected in their signed and written discourse. In connection with this, there is a need for reinvestigation of whether deaf (and some hard-of-hearing) readers do utilize a phonological strategy, or have established alternative, visually based mechanisms for successful reading. Like hearing readers, deaf individuals employ a number of cognitive strategies for decoding, comprehending, and recalling text. Because of their deafness, deaf readers may even have an advantage over hearing readers in the number of resources available to them. The fingerspelling system, inherent in the sign language system, may predispose deaf children to make the connection between expressive language and print at an earlier time than their hearing peers, when the connection is made explicit by parents and educators. Further, the deaf now have a number of technological tools that provide access to the Hearing world through print, which provides additional exposure to meaningful textual information. In sum, given the knowledge supplied here, there is no rational excuse for the failure of an intellectually average deaf individual from any background to achieve reading levels commensurate with those of their hearing, age-matched peers. Therefore, it is clear that in the case of reading and the deaf, the question should not be "Why can't Sam read?" but "Why shouldn't he read?" Why not, indeed?

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